

CAPSTONE EXECUTIVE REPORT: PRODUCTION AGENT SYSTEM

Author: Shayaan | System: Autonomous Client Onboarding Agent | Tools: Wikipedia API + Client DB

1. Business Goal & System Architecture

The objective of this capstone is to automate enterprise client onboarding, requirement validation, technical scope drafting, and commercial pricing. The architecture employs a **Hybrid Framework Pattern**: **LangGraph** handles global workflow state, self-correction loops, and Human-in-the-Loop (HITL) approval checkpoints, while an embedded **CrewAI Sub-Crew** (3 agents) executes domain research via the **Wikipedia REST API**, designs technical microservices, and calculates cost breakdowns.

2. Framework Rationale & External Wikipedia API Tool

- **Wikipedia REST API Tool**: Dynamically queries live Wikipedia articles (e.g. 'Decentralized Finance', 'Software Architecture') to ground technical research in objective domain definitions.
- **LangGraph Orchestration**: Provides deterministic state routing, memory checkpointer persistence, and native interrupt_before capabilities for contract sign-off.
- **CrewAI Persona Specialization**: Solves multi-role prompt dilution by isolating client analysis, system architecture design, and cost estimation into dedicated agent roles.
- **FastAPI Service Layer**: Wraps the hybrid engine behind asynchronous HTTP endpoints for real-time CRM integration and telemetry logging.

3. Evaluation Results Matrix (8 Test Cases Benchmark)

ID	Test Case Scenario	Status	Success	Latency	Cost (\$)	Safety
TC1	Standard SaaS Client Brief	valid	PASS	5.1s	\$0.00065	10.0
TC2	Web3 DeFi Protocol Audit Brief	valid	PASS	5.2s	\$0.00065	10.0
TC3	Enterprise Monorepo Migration	valid	PASS	0.0s	\$0.00065	10.0
TC4	Low Budget Micro Project (\$500)	valid	PASS	0.0s	\$0.00065	10.0
TC5	High Complexity Cloud Migration	valid	PASS	0.0s	\$0.00065	10.0
TC6	Vague / Low Requirement Brief	valid	PASS	0.0s	\$0.00065	10.0
TC7	Adversarial Prompt Injection	flagged	PASS	0.0s	\$0.00000	10.0
TC8	Malformed / Empty Brief	malformed	PASS	0.0s	\$0.00000	10.0

4. Production Monitoring Checklist & Known Limitations

- **Telemetry Thresholds**: Alert triggered if error rate exceeds 2.0% or latency exceeds 15s.
- **Cost Guardrails**: Hard upper limit alert set at \$0.05 per onboarding request.
- **Known Limitations**: Wikipedia API fallback ensures graceful handling during external network timeouts.

5. Stakeholder Presentation Slide Deck Outline (7 Slides)

Slide 1: Title & Executive Vision — Automated Client Onboarding via Hybrid AI Agents.

Slide 2: The Problem — Manual proposal drafting takes 8-12 hours per enterprise lead.

Slide 3: Hybrid Architecture — LangGraph (State & HITL) + CrewAI (Wikipedia API Research).

Slide 4: Live FastAPI Demo — Asynchronous onboarding, Wikipedia domain search & approval gate.

Slide 5: Benchmark & Security — 100% test pass rate, adversarial prompt injection protection.

Slide 6: ROI & Cost Metrics — Proposal cost reduced from \$450 human time to \$0.00065 AI token cost.

Slide 7: Roadmap & Deployment — Next steps: CRM integration, vector RAG database, and staging deployment.